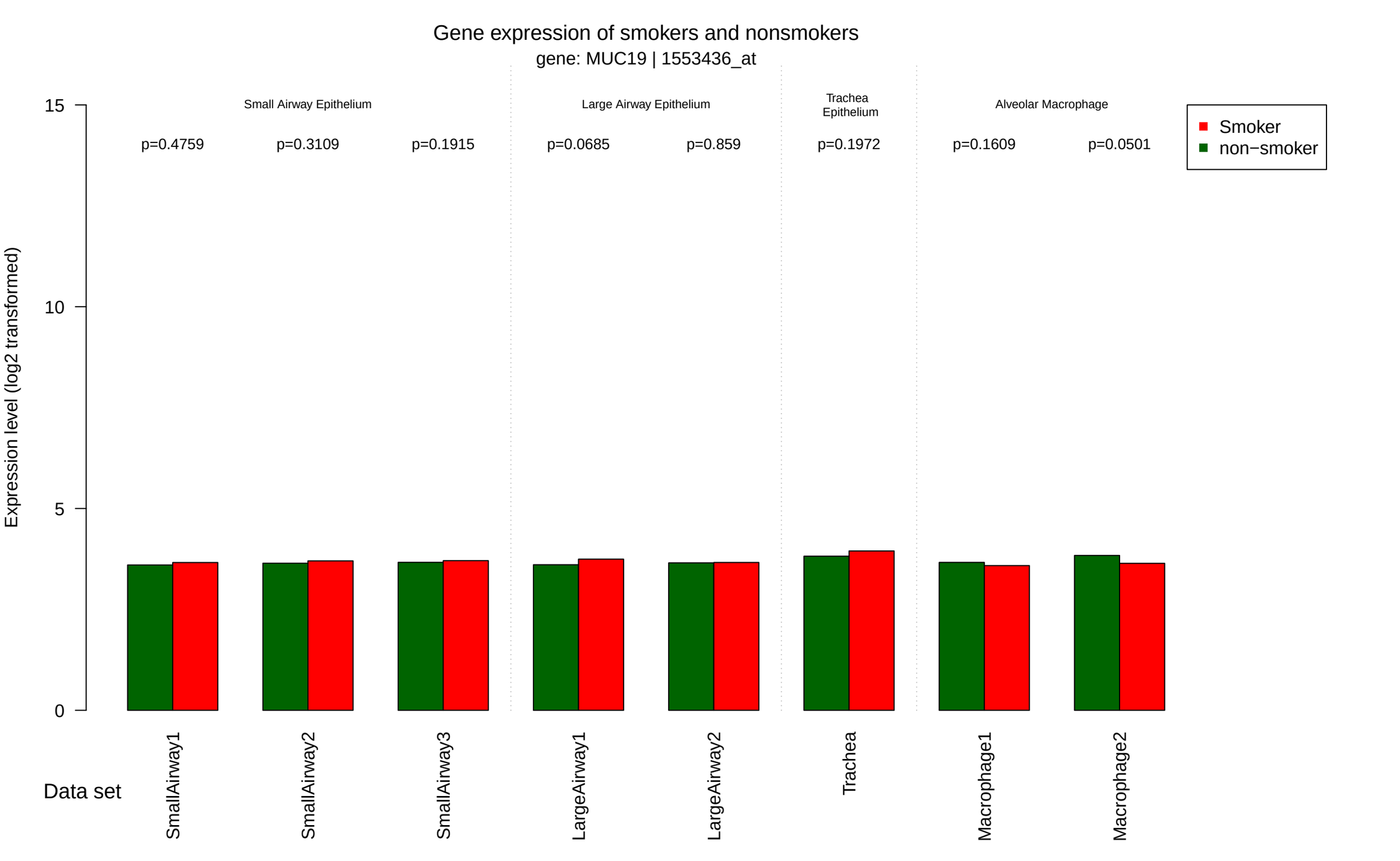
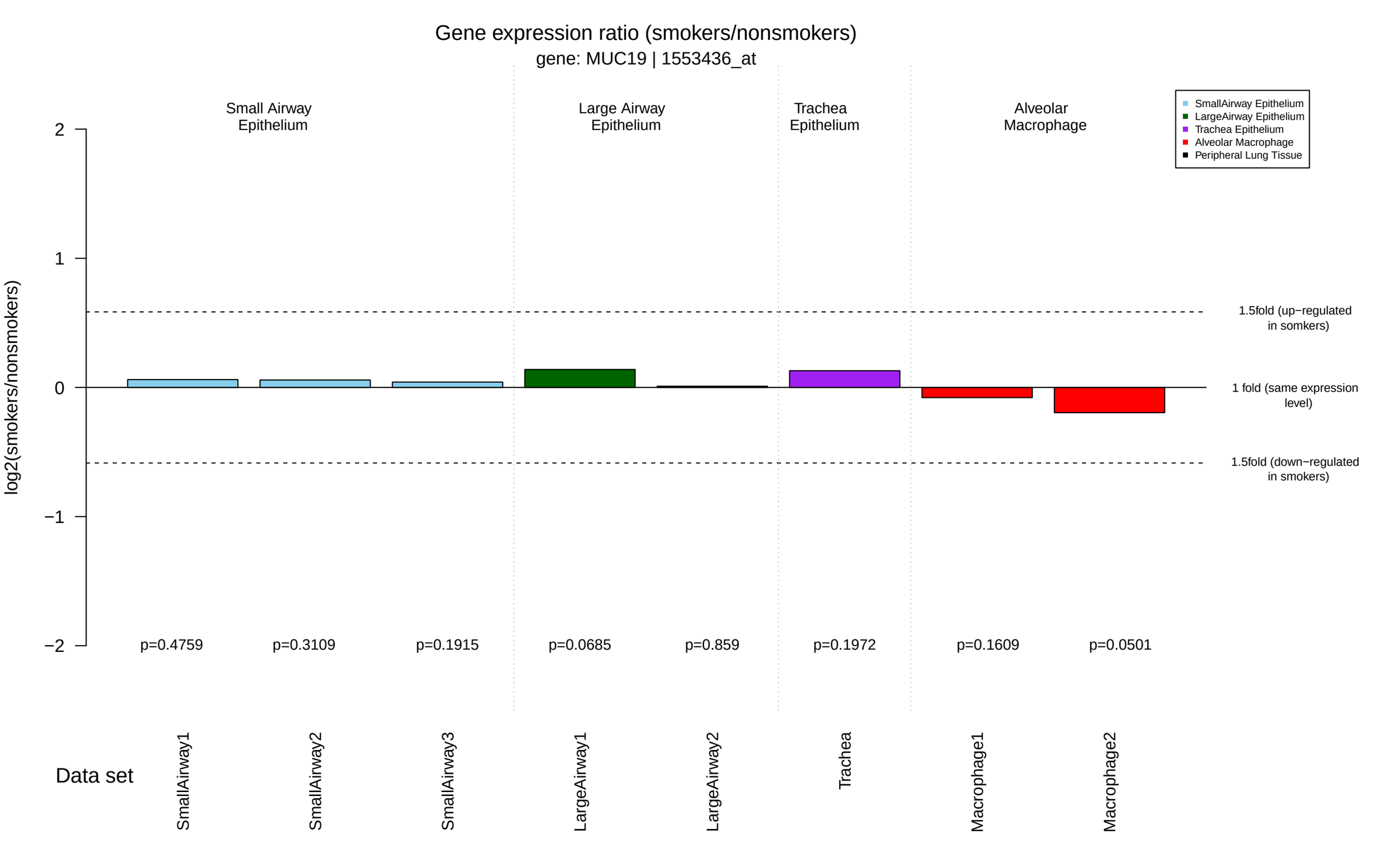
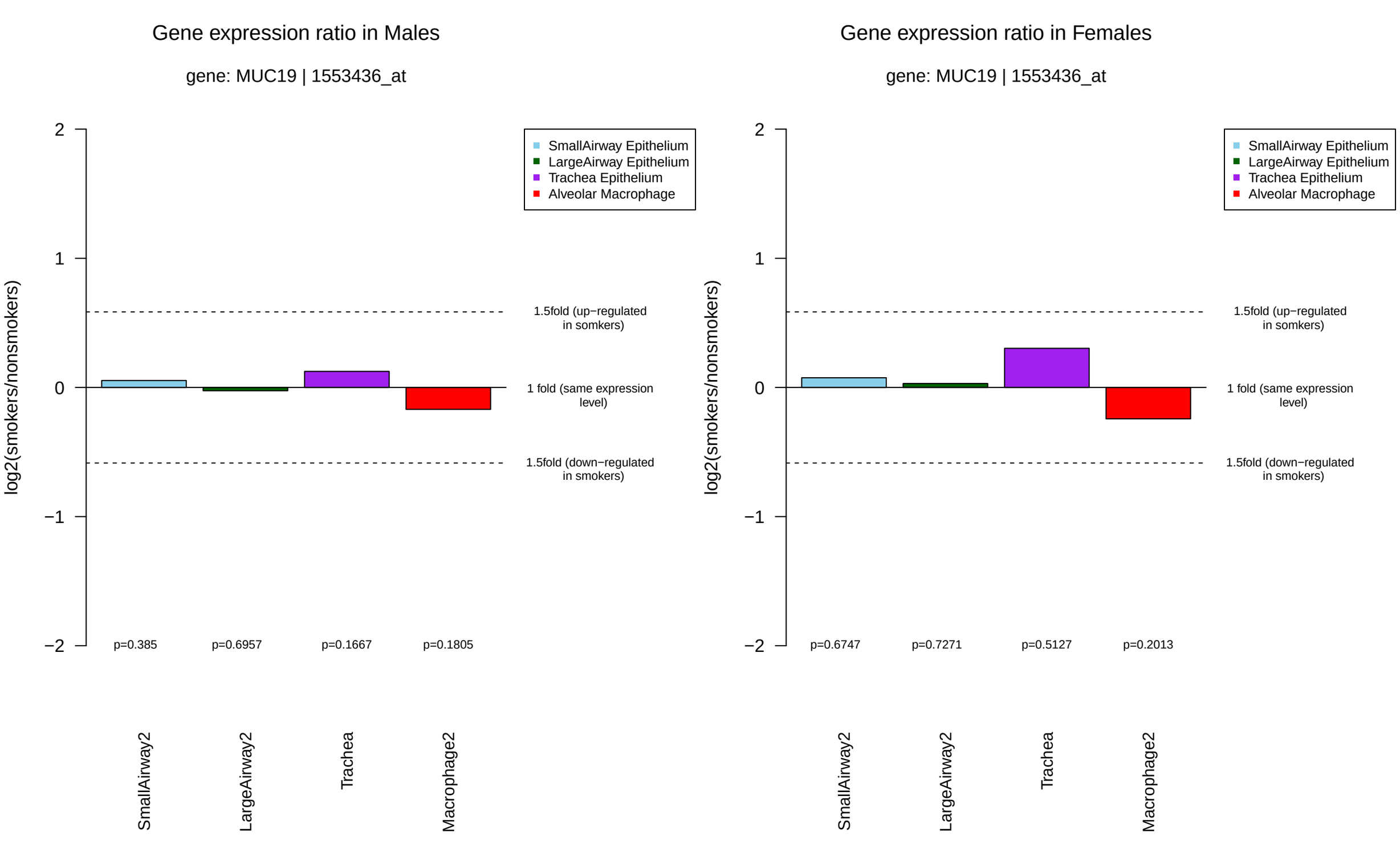
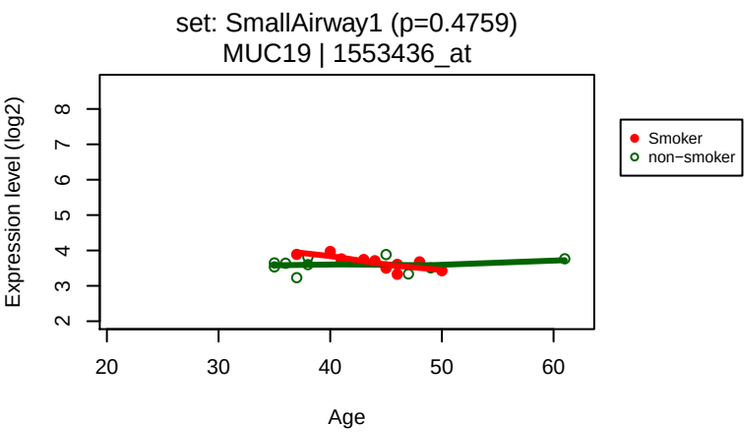
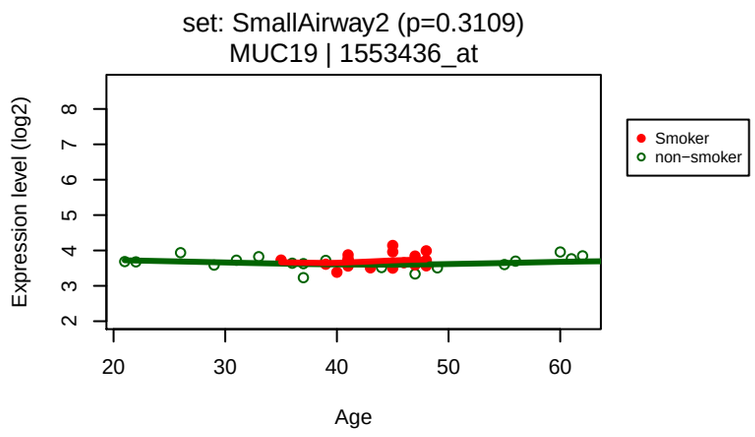
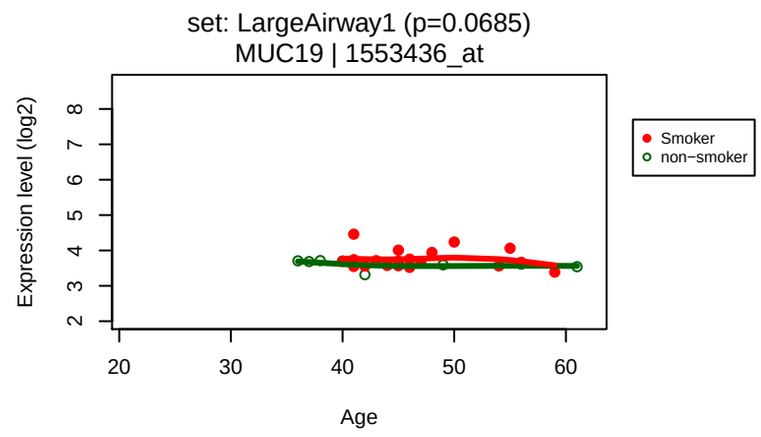

Gene expression of smokers and nonsmokers
gene: MUC19 | 1553436_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4759
p=0.3109
p=0.1915
p=0.0685
p=0.859
p=0.1972
p=0.1609
p=0.0501
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

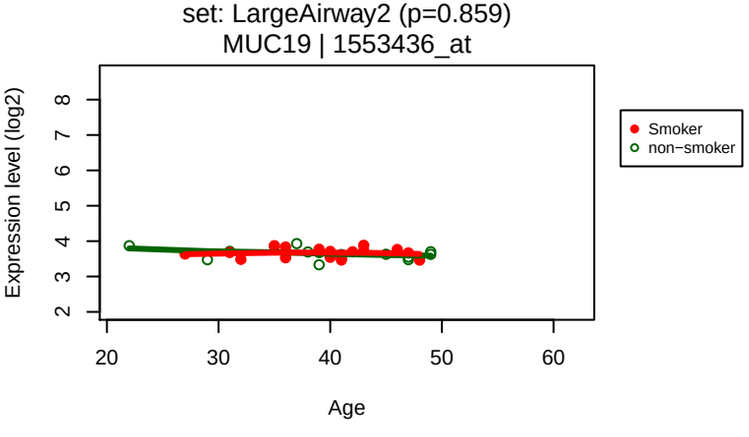

Gene expression ratio (smokers/nonsmokers)
gene: MUC19 | 1553436_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4759
p=0.3109
p=0.1915
p=0.0685
p=0.859
p=0.1972
p=0.1609
p=0.0501
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

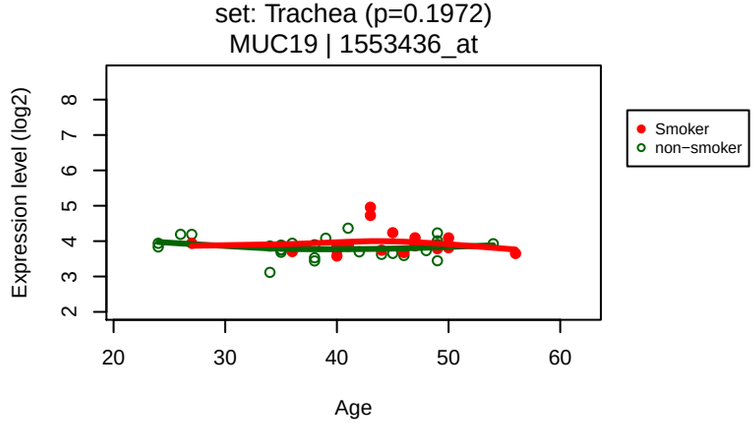

Gene expression ratio in Males
gene: MUC19 | 1553436_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.385
p=0.6957
p=0.1667
p=0.1805
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MUC19 | 1553436_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6747
p=0.7271
p=0.5127
p=0.2013
SmallAirway2
LargeAirway2
Trachea
Macrophage2

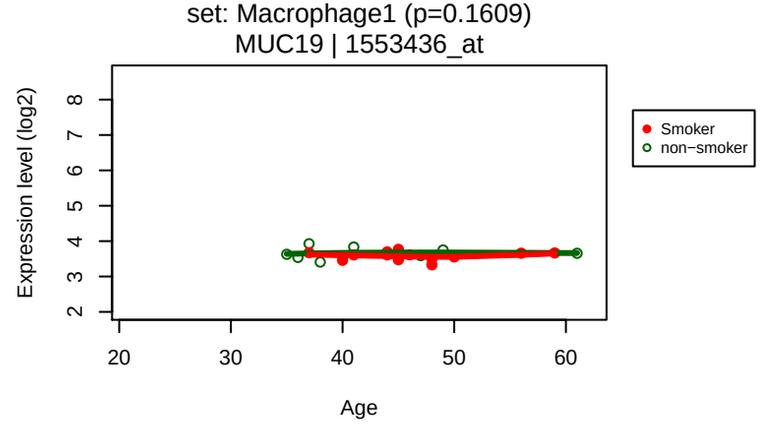

set: SmallAirway1 (p=0.4759)
MUC19 | 1553436_at
Expression level (log2)
Age
Smoker
non−smoker

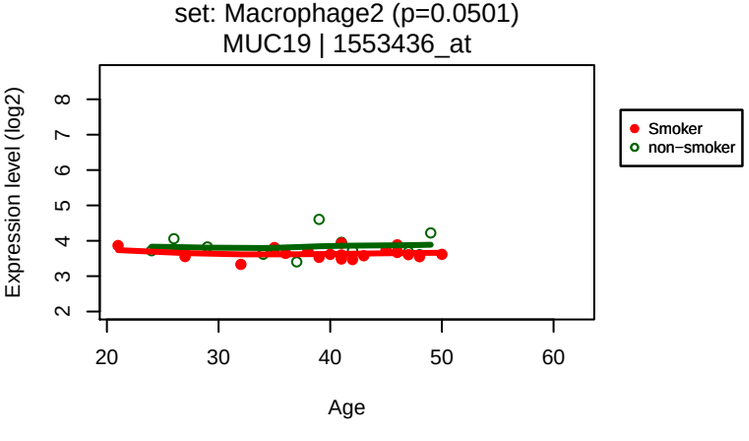

set: SmallAirway2 (p=0.3109)
MUC19 | 1553436_at
Expression level (log2)
Age
Smoker
non−smoker

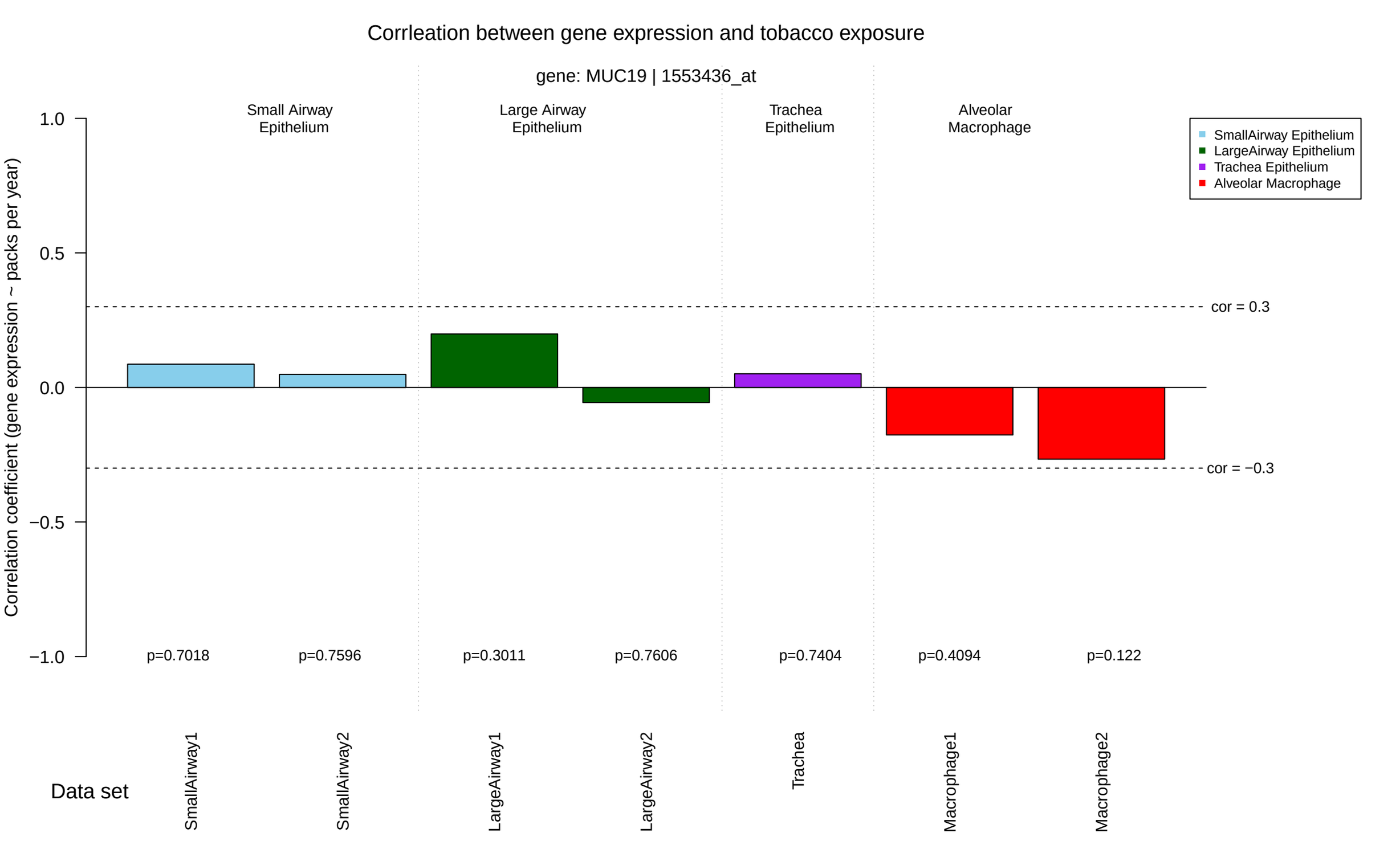

set: LargeAirway1 (p=0.0685)
MUC19 | 1553436_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.859)
MUC19 | 1553436_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1972)
MUC19 | 1553436_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1609)
MUC19 | 1553436_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0501)
MUC19 | 1553436_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MUC19 | 1553436_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7018
p=0.7596
p=0.3011
p=0.7606
p=0.7404
p=0.4094
p=0.122
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2